QP CODE 24900007

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCAQC100 - INTRODUCTION TO AQUACULTURE SYSTEMS AND RESOURCES

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer any 10 questions. each question carries 2 marks.

1. What is the main advantage of practicing Raceway Culture? [K] [1]

- Continuous water flow, ensures that fresh oxygenated water is continuously available
- Constant water flow, removes uneaten food and fish waste, preventing deterioration in water quality
- Higher fish yield and fish production possible

Any two of the above points or any other relevant points – 1 mark each for each point ; Total – 2 Marks

2. Name any two Stenohaline species? [K] [1]

Give the scientific names of any two freshwater or marine species, which cannot tolerate changes in salinity (eg; *Sardinella longiceps,* Indian major carps, minor carps, exotic carps , *Clarias batrachus, Hetropneustes fossilis, Channa* etc)

One mark each for each scientific name – Total: 2 Marks

3. Write the Scientific names of any two exotic carps? [K] [5]

1.*Cyprinus carpio* - Common Carp

2. *Hypophthalmichthys molitrix* – Silver carp
3.*Ctenopharyngodon idella* – Silver Carp
4. *Aristichthys nobilis* - Bighead Carp
5. *Mylopharyngodon piceus* – Black carp
6. *Cirrhinus molitorella Mud Carp*

Write any two scientific names correctly – 2 Marks (one mark each for each name)

4. Discuss the diurnal fluctuation of Oxygen in Culture ponds?

- During daytime dissolved oxygen concentration increases and reaches a maximum in the afternoon
- This is because oxygen is released during photosynthesis
- During night oxygen concentration decreases and declines to a minimum at dawn
- This is because , at night no photosynthesis occurs and oxygen is consumed only.

- Mention diurnal fluctuation and reason for the fluctuation ; 0.5 Marks each for each point – Total 2 Marks

5. What is meant by Soil Texture? [U] [2]

- Soil Texture refers to the relative proportions of sand , silt and clay present in the soil
- Soil containing 20-25 % clay, 65-70% sand and 15-20% silt are highly suitable for pond construction

First point – 1.5 Marks, 2nd point – 0.5 Marks ; Total – 2 Marks

6. Define Stocking Density? [U] [2]

- Stocking density is defined as the number of animals that can be stocked in unit area or volume of water

Definition – 2 Marks

7. Write any two benefits of Biofloc Technology [K] [3]

- Microbial activity controls ammonia levels reducing water pollution
- Fish or shrimp can feed on the biofloc reducing the need for commercial feed.
- Water Conservation – Minimal water exchange is done and hence less water is used than in traditional Aquaculture systems.
- By converting waste into a usable form, biofloc systems minimize environmental impact and promote recycling within the system.
- The microbial community maintained in the biofloc tanks can contribute to a healthier environment, potentially reducing disease and improving fish or shrimp growth rates.

Mention any two benefits – 2 Marks

8. What are the Primary components of an Aquaponics sytem? [K] [3]

1. Fish Tanks – for rearing Fish
2. Mechanical Filters – for removing solid wastes
3. Biological Filters – for removal of ammonia
4. Pump Tanks and Pumps – for pumping water for recirculation
5. UV-C light – for disinfection
6. Oxygenation devices, aeration devices – for increasing oxygen content of water.

Mention at least 4 components – 0.5 Marks for each component – Total -2 Marks

9. Comment on Pig –cum-Fish Culture? [U] [4]

-Integrated system of fish farming, where excreta and wastes of pig are used for feeding fish
- Pig Houses are constructed on pond embankments, waste and excreta of pigs are drained directly into the pond
- Pig dung is a good fertilizer causing heavy growth of phytoplankton, zooplankton and bottom detritus
- Fish also feed directly on pig excreta, which contains a good amount of digestible food for fish

Mention the concept of integrated farming – 1 mark
Explain any two benefits of pig culture for fish – 1 Mark
Total – 2 marks

10. Mention any two differences between Edible Oysters and Pearl Oysters? [A] [5]

	Edible Oyster	**Pearl Oyster**
1	Muscular foot absent	Presence of muscular foot
2	Byssus Threads absent, animals attached to substratum by a cement like secretion	Byssus threads present
3.	Shell valves unequal, left valve cup shaped	Shell valves unequal, circular in outline
4.	Outer surface of shell with sharp ridges	Finger like projections on ventral side of shell.
5.	Hinge short and not straight	Hinge line straight and presence of anterior and posterior ears
6.	Inner surface of shell not shiny	Inner surface of shell shiny, due to the presence of mother of pearl layer

Mention any two differences – 2 marks

11. Identify two seaweed species with high commercial value in aquaculture? [U] [5]

- *Gracilaria edulis, Gelidiella acerosa, Gracilaria crassa, Gracilaria verrucosa, Gigartina sp. , Chondrus crispus, Gigartina sp.Sargassum wightii, Turbinaria sp., Macrocystis sp. , Ascophyllum sp., Ulva lactuca, Porphyra sp., Laminaria sp., Undaria sp. Etc.*

Give the genus name for any two species – 2 Marks

12. Mention any two species of cultivable bivalves with their scientific names? [K] [5]

Villorita cyprinoides, Anadara granosa, Marcia opima, Meretrix meretrix, Meretrix casta, Paphia malabarica, Crassostrea madrasensis, Crassostrea gryphoides Perna viridis, Perna indica, Pinctada fucata, Pinctada margaritifera

Mention the scientific names of any two species – 2 Marks (one mark each for each name)

Part B
Short Answer Questions
Answer 3 questions.
Each question carries 6 marks

13 Elaborate upon the significance of Ammonia in Fish culture ponds and its Management [U] [2]

- Originates in Aquatic ecosystems through excretion by fish
- Extremely toxic to fish

Harmful effects

- Ammonia occurs in two forms**,** unionized ammonia and ammonium ion
- As ammonia concentration in water increases , excretion of ammonia by aquatic animals decrease
- Ammonia concentration in blood increases, causes increases in blood pH, affecting metabolism
- Ammonia increases oxygen consumption by tissues, damages gills and reduces ability of blood to transport oxygen

Management

- Maintain sufficient level of dissolved oxygen facilitates oxidation of ammonia
- Application of zeolite to absorb ammonia from bottom sediments

Origin of ammonia in culture ponds – 1 Mark
Harmful effects of ammonia – 3 Marks
Management of ammonia – 2 Marks
Total – 6 Marks

14 Explain the major benefits and challenges of recirculatory aquaculture System [K] [3]

Advantages of Recirculating Aquaculture System

- **Reduced Water Usage:** RAS significantly reduces water consumption by recirculating and treating water.
- **Minimised Pollution:** By effectively filtering and treating water, RAS minimizes the release of pollutants into the environment
- **Efficient land Use:** RAS can be operated in smaller spaces, making it suitable for urban and rural areas
- **Reduced risk of diseases:** The controlled environment in RAS helps prevent the spread of diseases.
-**Production duration, Quantity and Quality:** RAS allows for year round production, regardless of seasonal variations.
-RAS can support higher fish densities leading to increased production in a smaller space
- Controlled water conditions and precise feeding strategies result in higher-quality fish.
- **Precise water quality Control:** RAS allows for precise control of water parameters such as temperature, pH, dissolved oxygen, and nutrient levels.
- **Easy Grading and Harvesting of Fish:** The controlled environment of RAS allows for consistent and predictable growth, simplifying the grading process.

CONSTRAINTS IN RAS SYSTEMS

1. Necessity of Electricity continuously, without fail
2. Good water source
3. Good fish feed quality, preferably high protein and fat extruded diets with high digestibility.
4.Technically skilled staff.

Benefits – Any four points – 4 marks
Challenges – Any two points- 2 Marks
Total – 6 Marks

15 Discuss the role of Duck-cum-Fish Culture in improving water quality and fish productivity [U] [4]

- Duck droppings is a good fertilizer , rich in carbon, nitrogen, potassium, phosphorus, calcium etc.
- Dabbling of ducks at the bottom releases nutrients from soil
- They feed on mollusks , insect larvae, tadpoles and aquatic weeds.
- Ducks consume excessive algae, preventing algal blooms
- Ducks swimming in the pond helps to aerate water, increasing oxygen levels in the pond
- Ducks feed on leftover food, weeds and insects around the pond , minimizing waste and reducing risk of water pollution from uneaten organic matter.
- Duck droppings contains beneficial microbes that help break down organic matter, improving water clarity and quality.
- Nutrients in duck droppings stimulate growth of plankton, which is feed for fishes.

Any 4 points (1.5 marks each for each point) – Total 6 Marks

16. Describe the morphology of a penaeid prawn [K] [5]

- Body divisions, Cephalothorax , abdomen, Symmetry
- Exoskeleton, Carapace, Tergum, Sternum, Pleuron, Rostrum
- Description of typical biramous appendage and other appendages
- External openings of the body
- Sexual dimorphism.

All of the above points covered with descriptions – 6 Marks
Reduce marks proportionately for points not covered

17 Describe the economic importance of sea urchins [U] [5]

a. **Sea Urchin Roe:**
- considered as a delicacy by Japanese
- Roe obtained from *Pseudocentrus depressus* , *Hemicentrotus pulcherrins* etc
- Eggs processed within 4-5 hours of collection, shell is removed and eggs placed on a draining surface, washed in ice water before packing, frozen before shipping.; also salted and preserved.
- Eggs eaten raw with lemon juice and soy sauce
- Fermented eggs are eaten with rice

b. **Curios**
- Hemispherical or globular shaped shell
- Used as decorative items or ash trays.

Explanation on sea urchin roe – 4 marks; Explanation on Curios – 2 Marks; Total - 6 Marks

Part C
Essay Type Questions
Answer any 1 question.
Each question carries 12 marks

18 Compare the extensive, semi-intensive and intensive methods of Aquaculture [U] [1]

	Extensive	Semi- Intensive	Intensive
Pond Size	1-5 hectares	0.2 – 2 hectares	0.1 – 0.5 hectares
Production levels	Varies from 50-700 kg/ha/year	500-5000 kg/ha/year	10000 – 15000 kg/ha/year
Stocking	* Seed carried into ponds by tides along with undesirable species * Polyculture commonly practiced	* Seed may enter during tides, but mostly desired species of same size are obtained from hatcheries * Use of pesticides and toxicants to control predators	* Only hatchery produced seed are used, all of the same size * Use of toxicants like teaseed cake, Rotenone to control predators

Feed	Natural production with organic manures and wastes.	Natural production during early growth stages, but formulated feed is given in the later stages.	Only high quality and costly formulated feed is given
Water exchange	Only tidal exchanges of water at levels of 5% or less	20-50 % of water exchange by tides and pumps.	100-300 % of water exchange daily by pumps
Aeration	No aeration devices used	Aeration used for agitating water	Aeration is given continuously using aerators and blowers

Elaborate upon any 6 differences between the 3 types of culture systems – 2 marks each for the explanation of each point of difference – Total – 12 Marks

19 Summarise the different Physical and Biological water quality parameters in Fish culture and their management [U] [2]

Physical Factors influencing water quality
1. Temperature 2. Turbidity 3.Light 4. Water Depth
Biological Factors influencing Water Quality
1. Phytoplankton 2. Aquatic Plants 3. Presence of Predatory and other fishes

Explain at least 5 parameters
Points to be elaborated upon for each parameter- 1. Influence of the parameter on water quality and cultured animals 2.Measurement of the parameter if possible 3. Management of the parameter.
Explanation of any 5 parameters – 12 Marks, any 4 – 10 marks, any 3 – 8 Marks, any 2 – 6 marks, any one only – 4 marks

20 Write an essay on cultivable seaweed and their economic importance [U] [5]

Mention cultivable seaweeds from
Red algae- *Gracilaria, Gelidiella, Porphyra, Hypnea , Acanthophora* etc.
Brown Algae – *Laminaria, Undaria , Sargassum, Macrocystis, Ascophyllum* etc.
Green algae- *Ulva lactuca, Caulerpa racemosa* etc.
Economic uses: Manufacture of agar, agaroids, alginates, as animal feed, fertlisers and manures, for human consumption etc.

Explain in detail the seaweed and its economic importance, any 6 seaweeds and their uses , explanation in detail needed- 6x2 = 12 Marks